

1

2

3

4

A

B

C

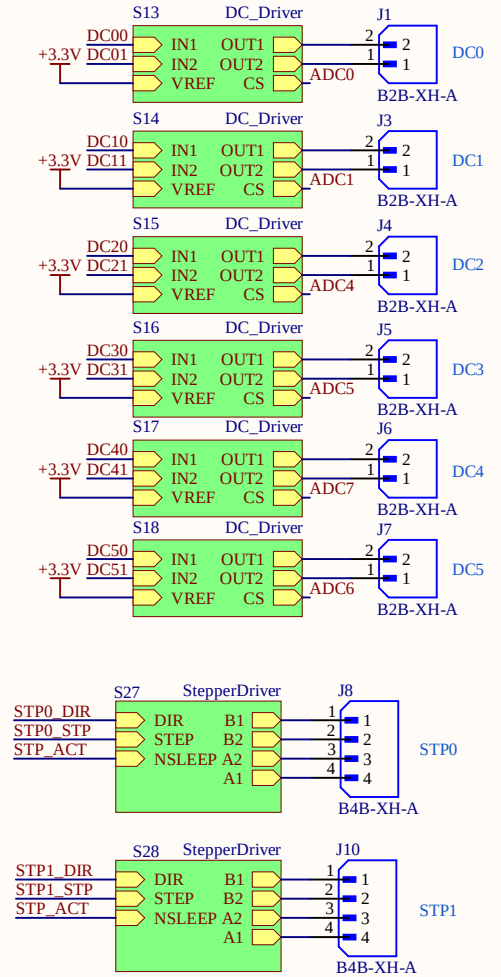
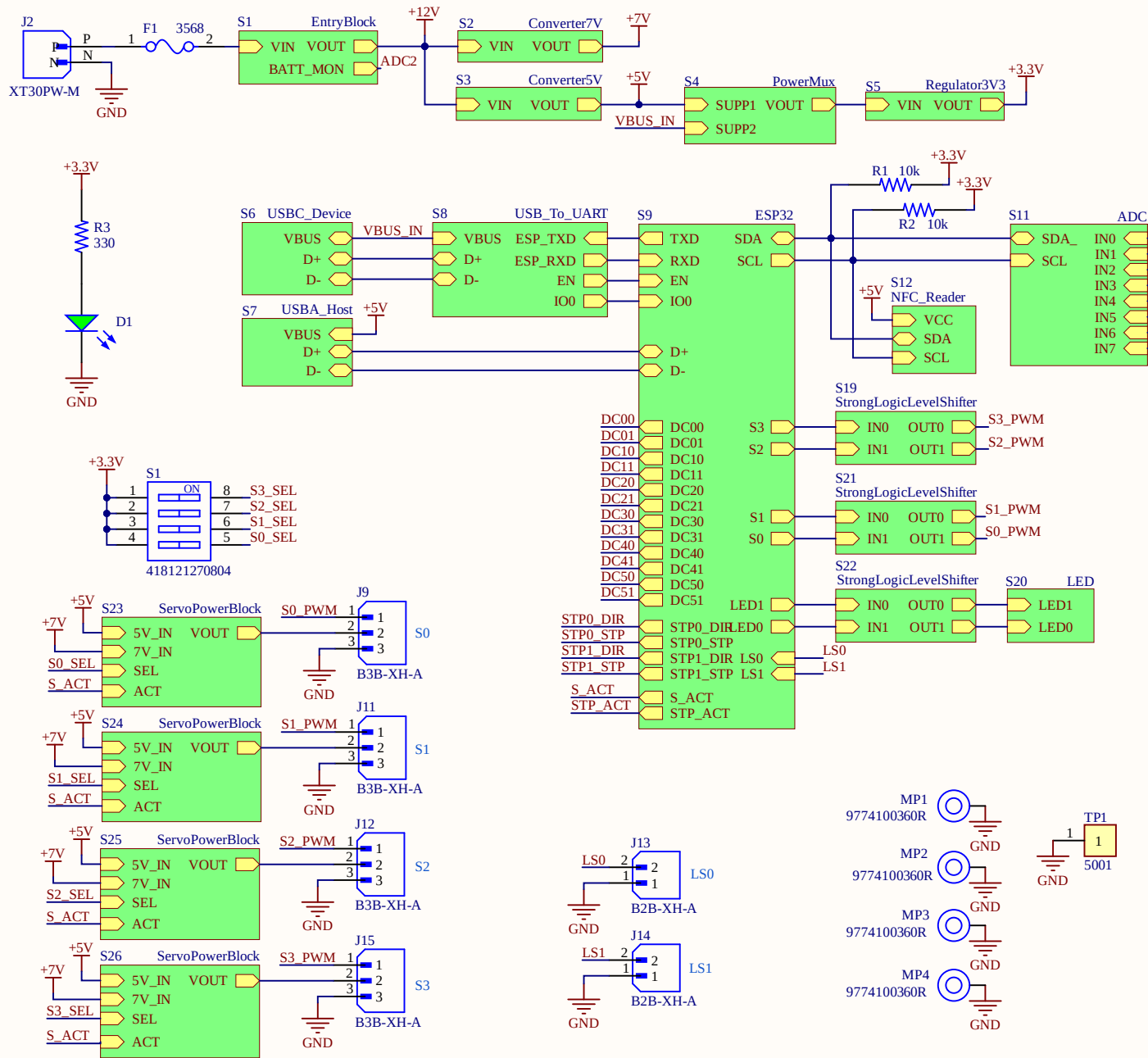
D

A

B

C

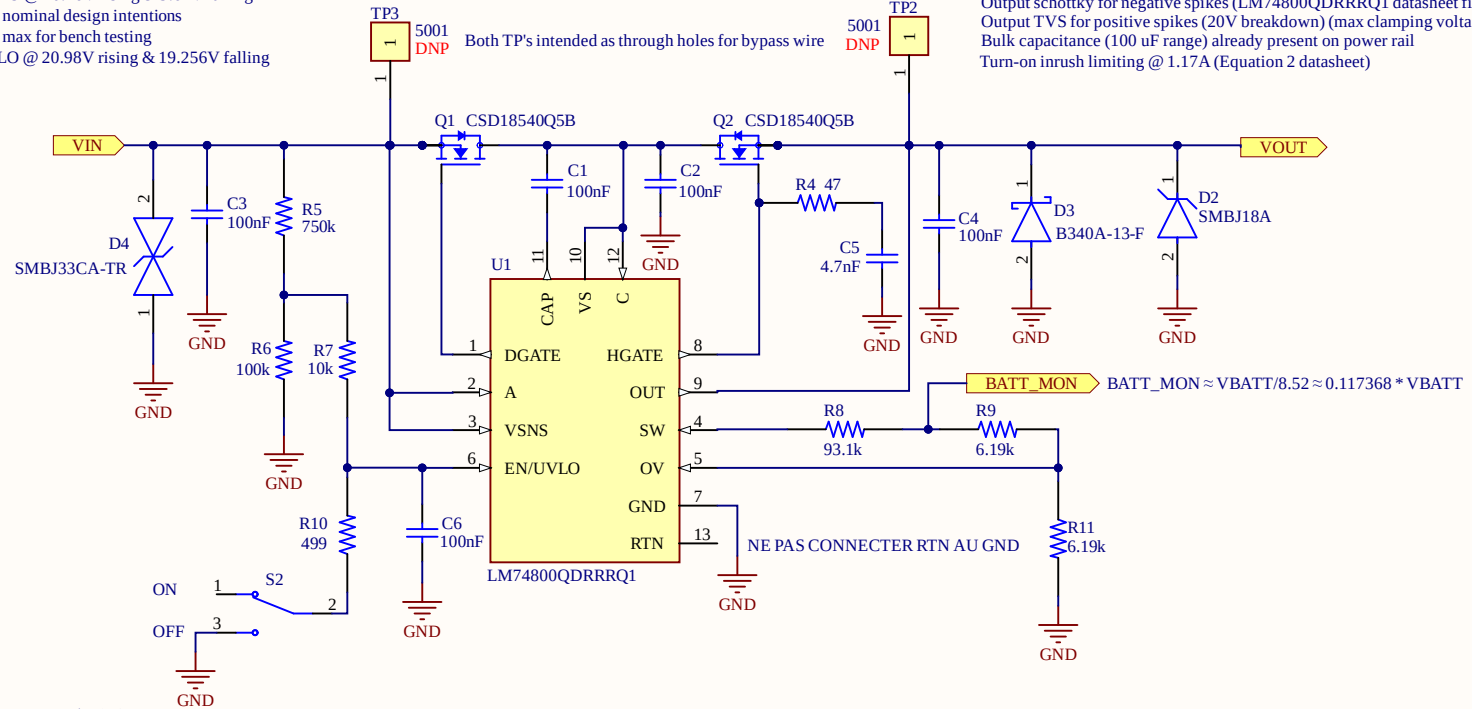
D



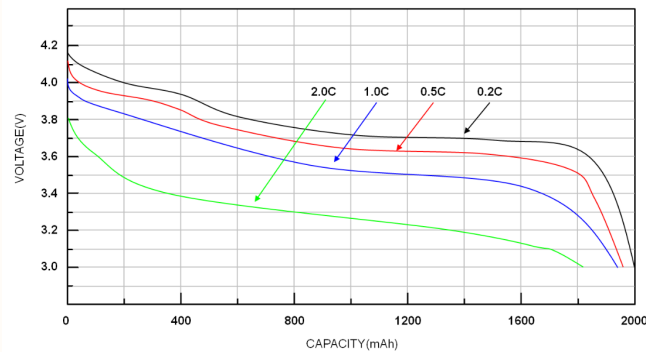
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IC has reverse input protection to -65V and 65V positive input range
 Input TVS for positive and negative spikes (36V standoff, 58.1V clamping)
 UVLO @ 10.46V rising & 9.62V falling
 12V nominal design intentions
 18V max for bench testing
 OVLO @ 20.98V rising & 19.256V falling

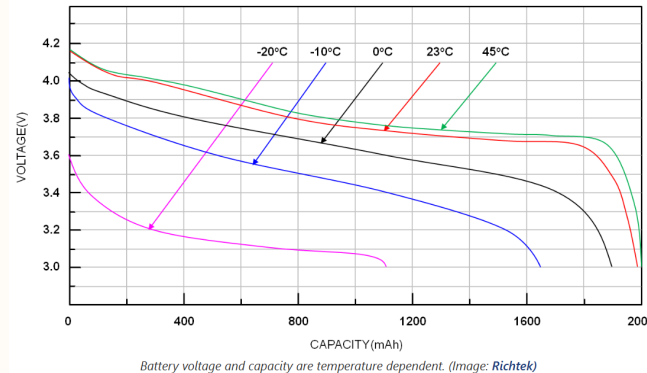
Downstream IC's damaged at 28V
 Downstream capacitors rated for 25V
 Output schottky for negative spikes (LM74800QDRRRQ1 datasheet fig. 11-1)
 Output TVS for positive spikes (20V breakdown) (max clamping voltage at 29.2V)
 Bulk capacitance (100 uF range) already present on power rail
 Turn-on inrush limiting @ 1.17A (Equation 2 datasheet)



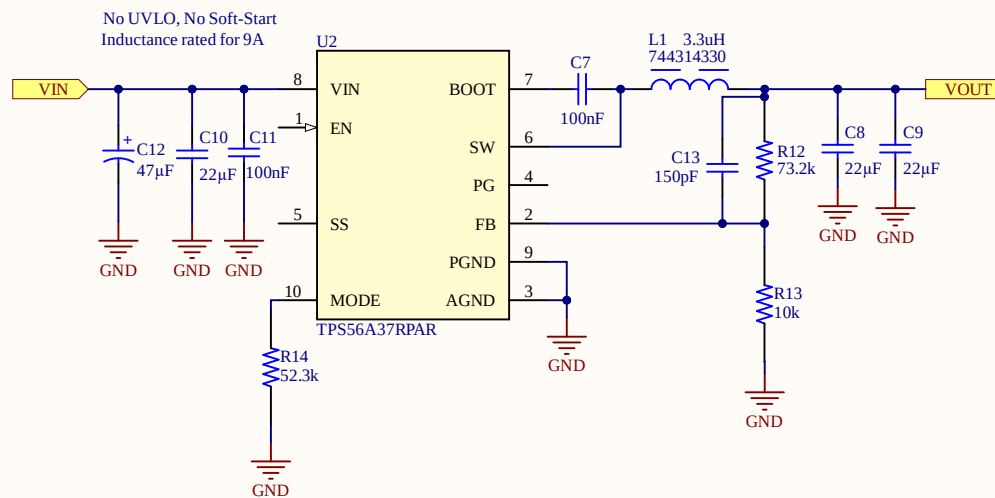
On utilise une batterie Li-Ion (ou LiPo) 3S
 Donc faire x3 sur les valeurs des graphiques



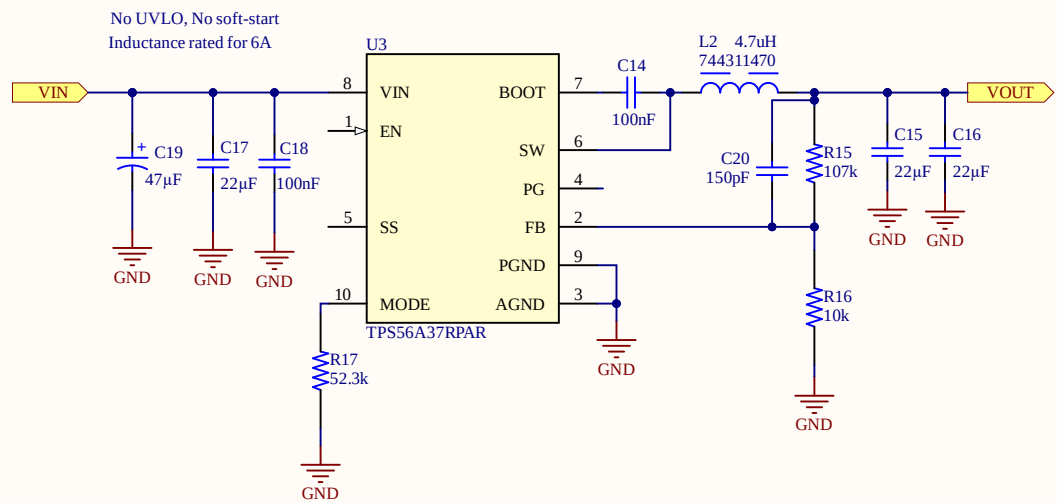
UVLO @ 3.49V rising & 3.21V falling (3S battery) Full battery : 12.6V
 OVLO NA for 3S battery Empty battery : 9V



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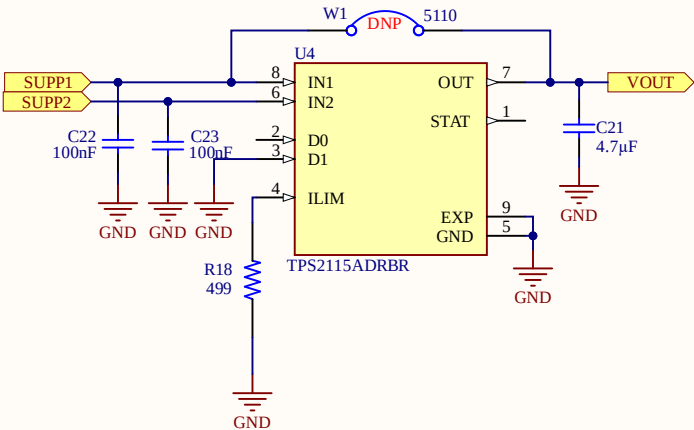


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File:	C:\Users\...\Converter7V.SchDoc	Drawn By:

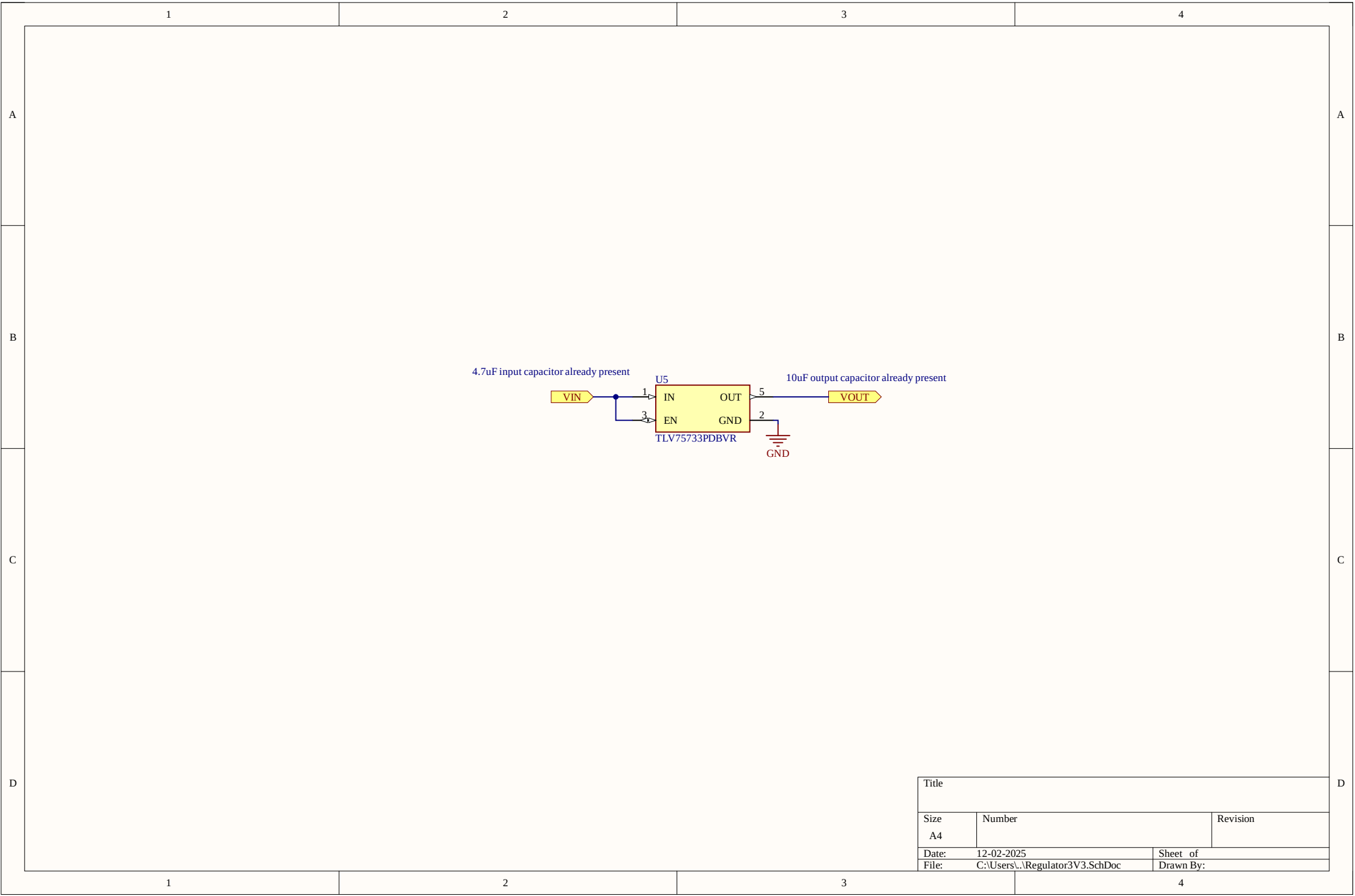
D0 is pulled-up internally, D1 shorted to GND

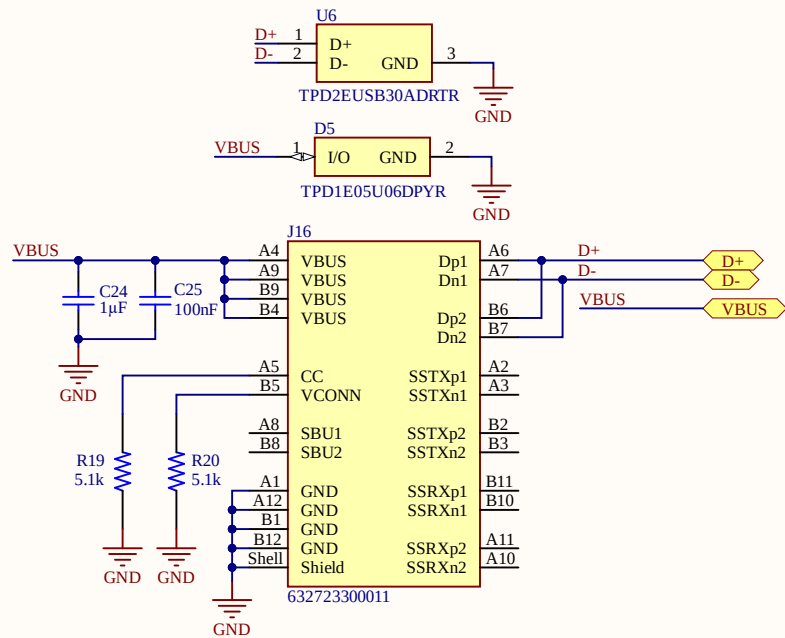
D1	D0	$V_{I(IN2)} > V_{I(IN1)}$	STAT	OUT ⁽¹⁾
0	0	X ⁽²⁾	Hi-Z	IN2
0	1	No	0	IN1
0	1	Yes	Hi-Z	IN2
1	0	X	0	IN1
1	1	X	0	Hi-Z

Current limit set at 1A (ILIM pin) (Only powers the 3.3V rail through the LDO)
Output capacitor to reduce voltage droop at switchover (see datasheet fig. 5)



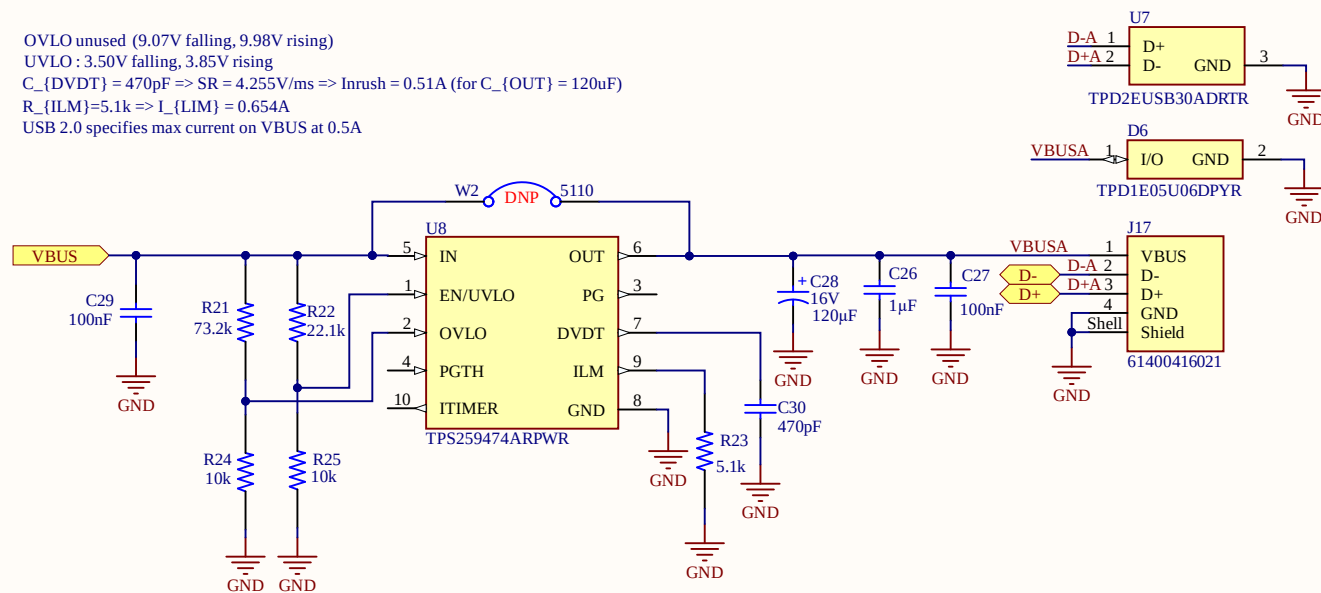
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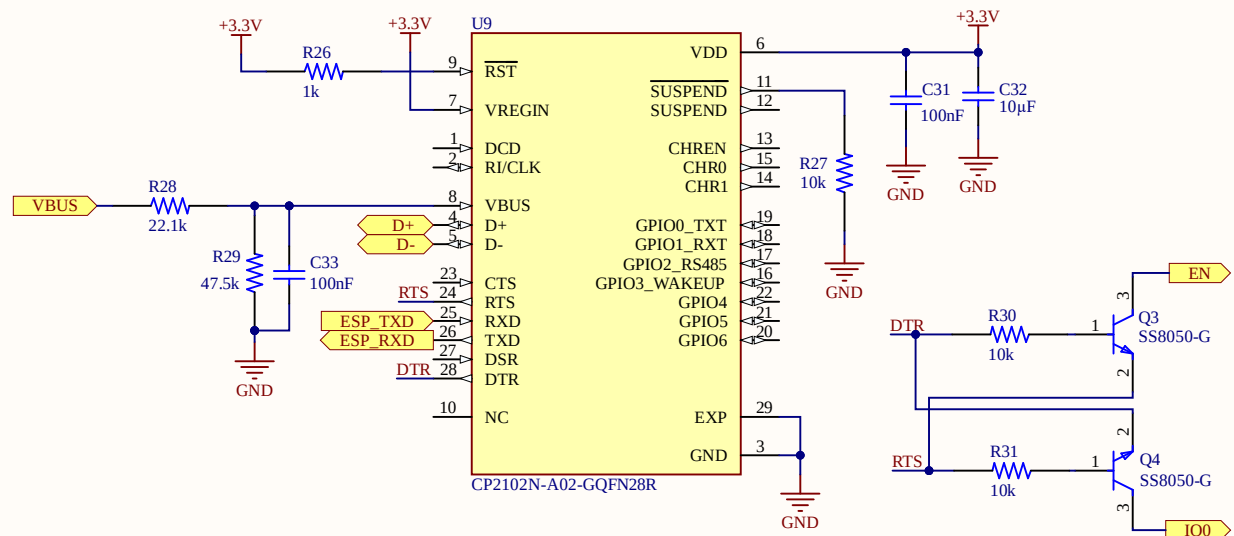


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OVLO unused (9.07V falling, 9.98V rising)
 UVLO : 3.50V falling, 3.85V rising
 $C_{\{D\}VDT} = 470\text{pF} \Rightarrow SR = 4.255\text{V/ms} \Rightarrow \text{Inrush} = 0.51\text{A}$ (for $C_{\{OUT\}} = 120\mu\text{F}$)
 $R_{\{ILM\}} = 5.1\text{k} \Rightarrow I_{\{LIM\}} = 0.654\text{A}$
 USB 2.0 specifies max current on VBUS at 0.5A

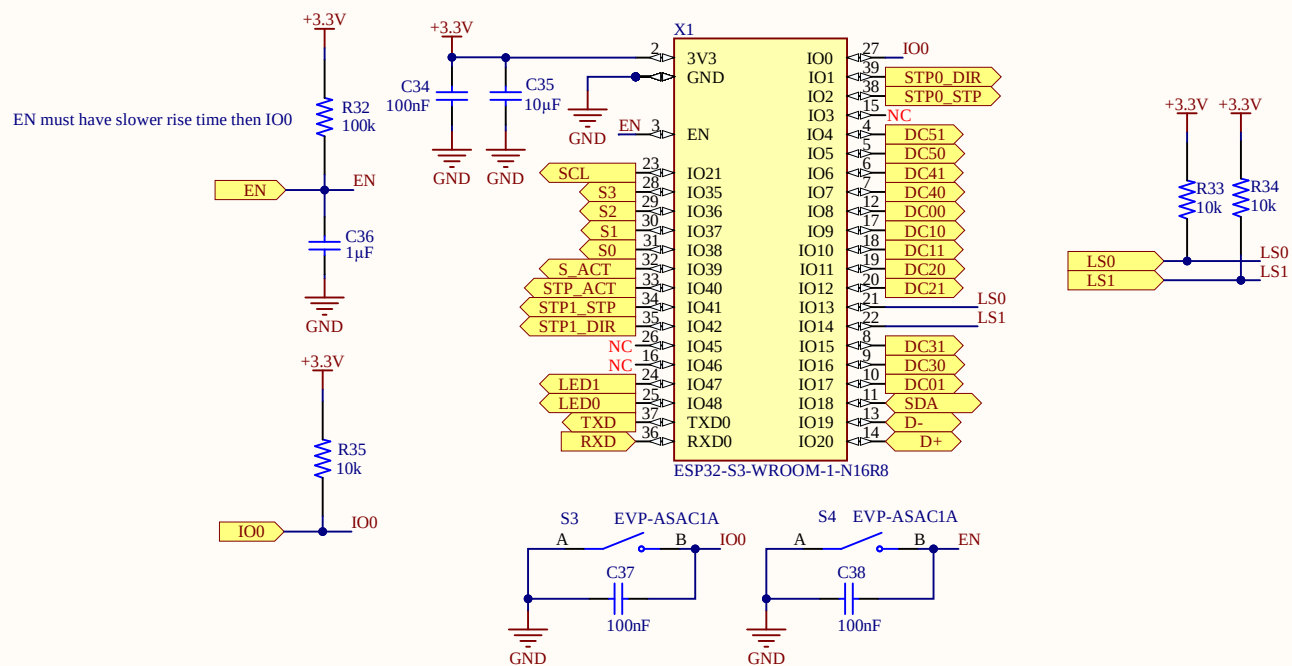


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SOURCE
https://dl.espressif.com/dl/schematics/SCH_ESP32-S3-DevKitC-1_V1.1_20220413.pdf

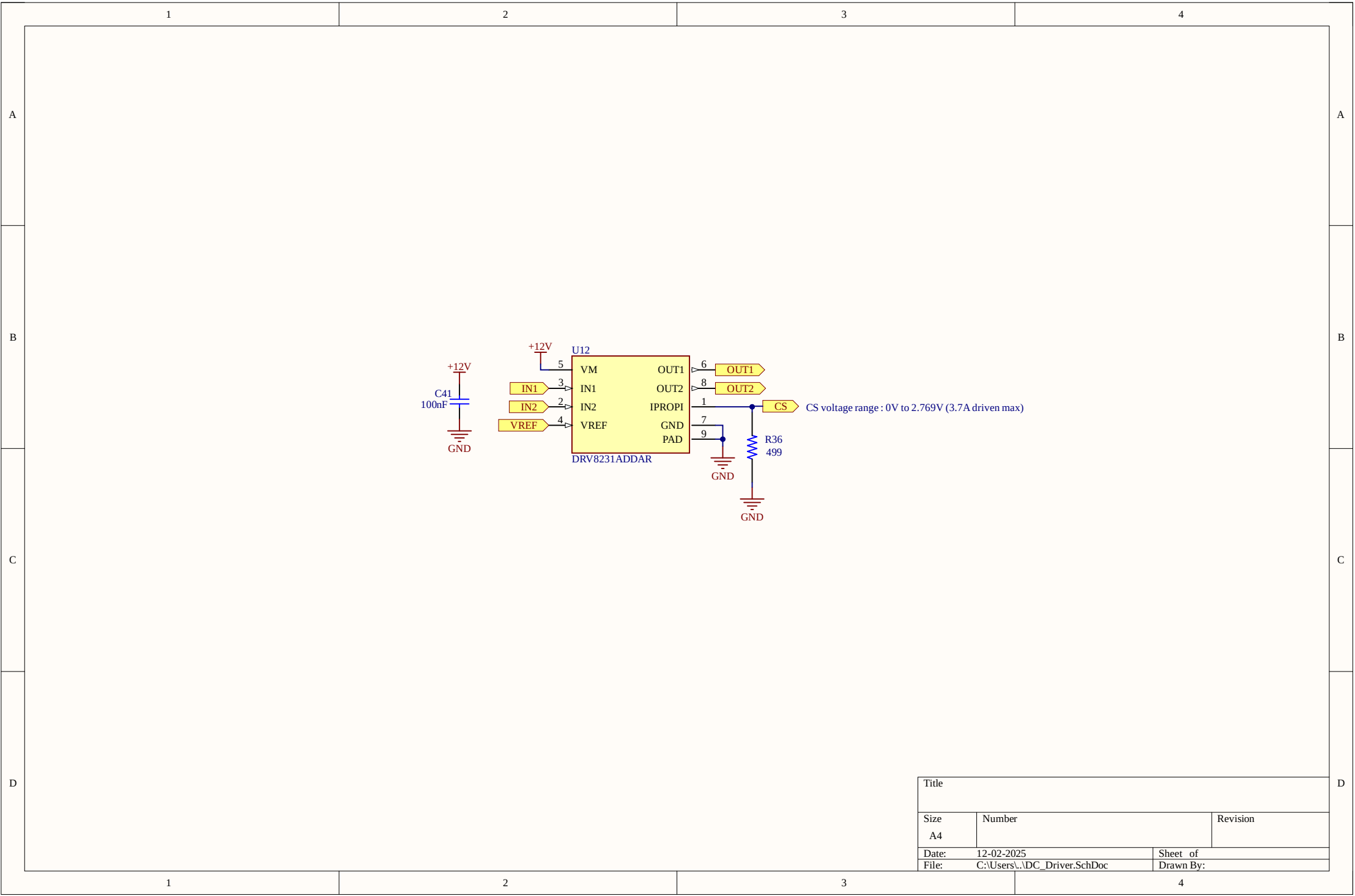
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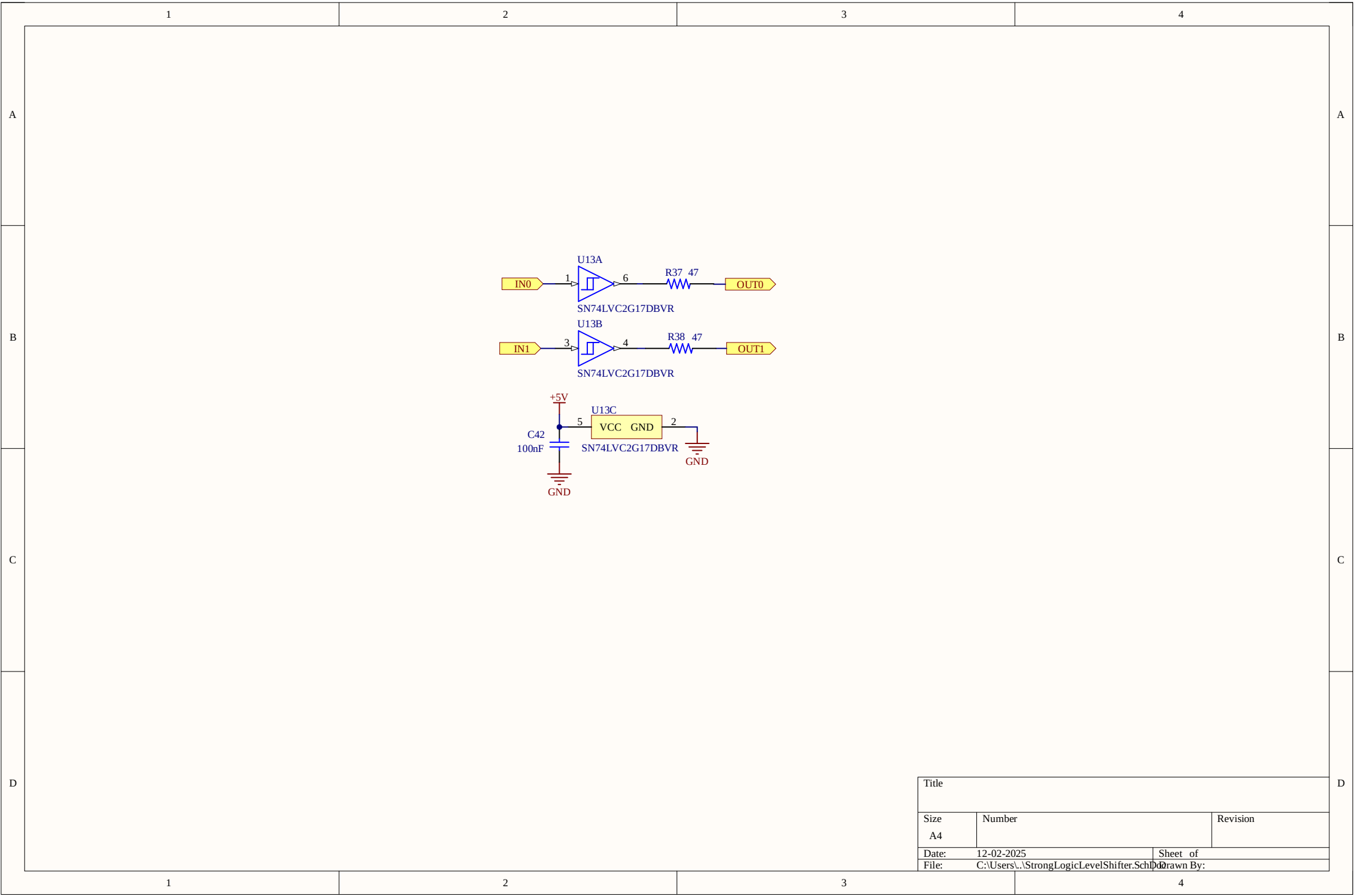


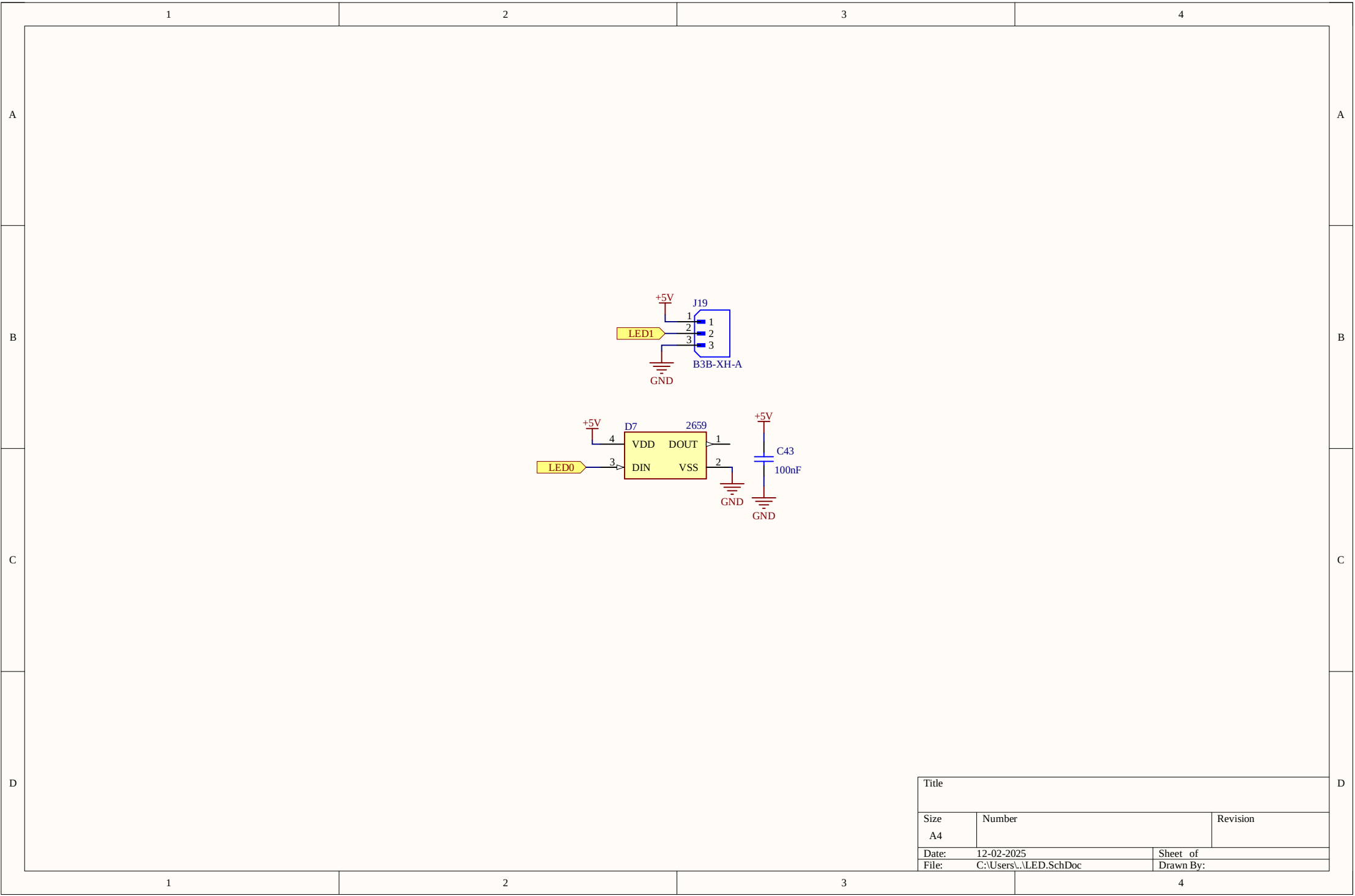
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https://dl.espressif.com/dl/schematics/SCH_ESP32-S3-DevKitC-1_V1.1_20220413.pdf

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SERVOMOTEUR 25kg :
<https://www.aliexpress.com/item/1005005073143851.html>
 Stall current 3.2A
 On set notre current limit à 3.33A (ILIM pin)
 On laisse la pin ITIMER floating pour le fastest overcurrent response
 Output capacitance : ~70uF, dVdt capacitor : 470pF, output slew rate : 4.25V/ms, inrush current : 0.298A
 OVLO unused (9.07V falling, 9.98V rising)

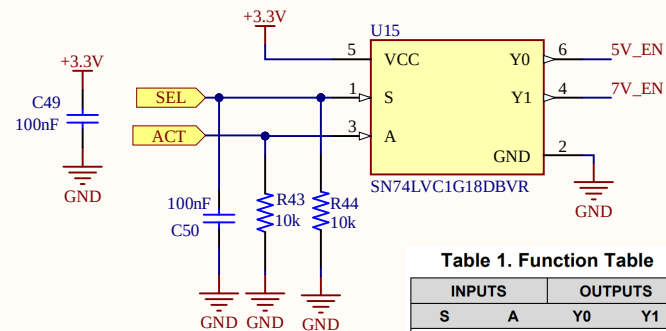
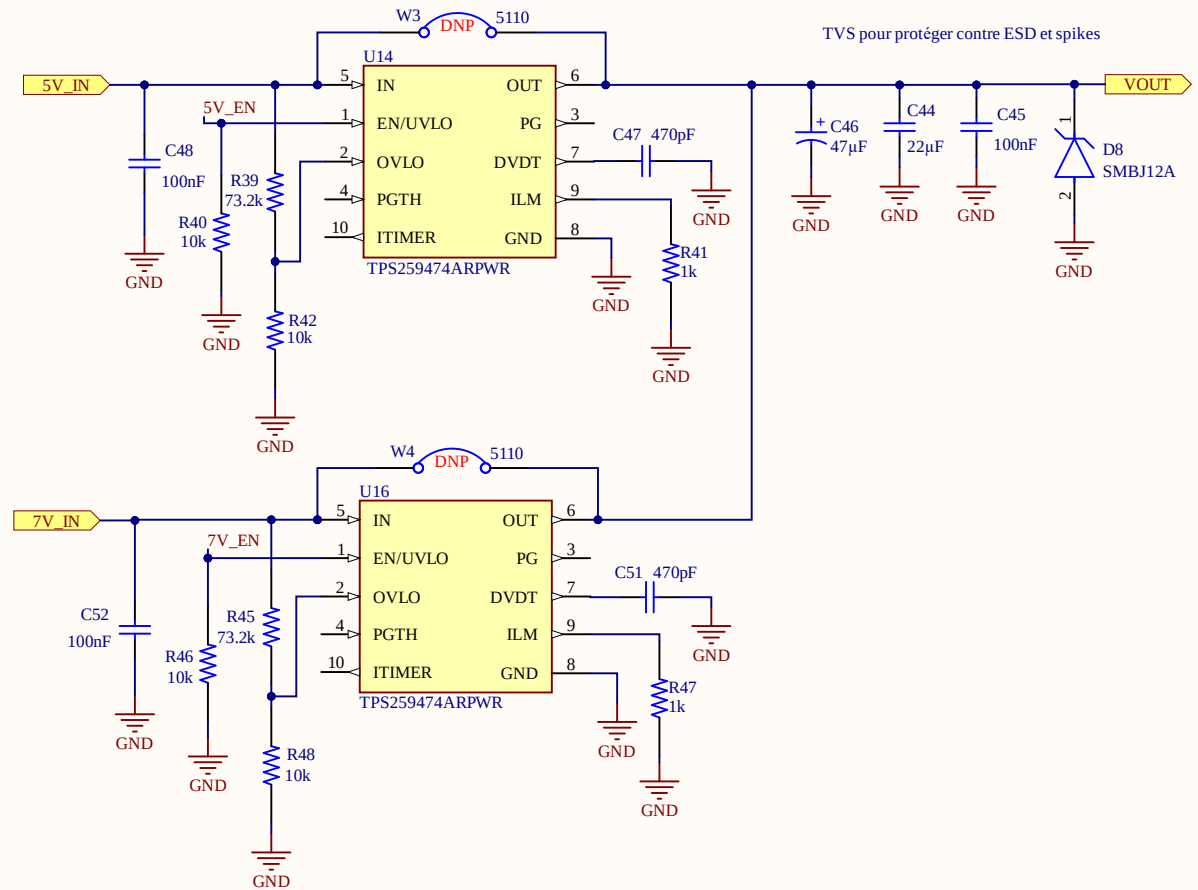
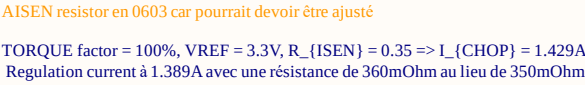


Table 1. Function Table

INPUTS		OUTPUTS	
S	A	Y0	Y1
L	L	L	Z
L	H	H	Z
H	L	Z	L
H	H	Z	H



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